IPv6 (Part-3)

- Topics covered:
  - A Correction
  - IPv6 header
  - Neighbour Discovery Protocol (NDP)
  - SLAAC
  - IPv6 Static Routing

A Correction

Correction About IPv6 Address Representation

The correction relates to how IPv6 addresses are written.

While researching IPv6, RFC documents were reviewed.

What is an RFC?

RFC means Request for Comments.
RFCs are published by:

- ISOC (Internet Society)
- IETF (Internet Engineering Task Force)

RFCs define official standards for:

- Internet protocols
- Specifications
- Procedures

Example: OSPF has many RFCs explaining how it works.

RFC 5952 — IPv6 Address Text Representation

RFC 5952 recommends standardizing IPv6 address formatting.

Previously, IPv6 writing was flexible:

- Leading zeros could be kept or removed
- Zero quartets could optionally be shortened
- Uppercase or lowercase hex could be used

RFC 5952 suggests one consistent format.

Rules from RFC 5952

1. Leading zeros MUST be removed

IPv6 quartets must not contain unnecessary zeros.

2. Double colon (::) MUST shorten the longest zero sequence

- If one zero quartet only → do NOT use ::
- If multiple zero blocks exist → shorten the longest

3. If equal-length zero blocks exist

→ Shorten the LEFT one

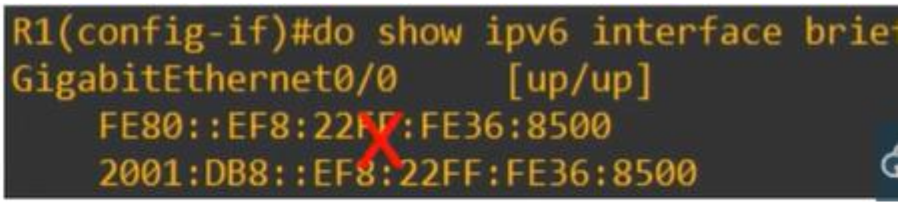
4. Hex letters MUST be lowercase

Letters a–f must be lowercase, not uppercase.

Cisco Devices and RFC Rule

Cisco routers sometimes display uppercase hex characters.
This is technically incorrect per RFC, but it is not a major issue.

The recommendation is to follow lowercase standard going forward.



IPv6 Header

All About IPv6 Header

IPv6 header is simpler than IPv4.

Comparison with IPv4

IPv4 header:

- Variable size (20–60 bytes)

IPv4 header format

Offsets	Octet	0								1								2								3							
Octet	Bit	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
0	0	Version				IHL				DSCP						ECN		Total Length															
4	32	Identification																Flags			Fragment Offset												
8	64	Time To Live								Protocol								Header Checksum															
12	96	Source IP Address																															
16	128	Destination IP Address																															
20	160	Options (if IHL > 5)																															
24	192																																
28	224																																
32	256																																

IPv6 header:

- Fixed size = 40 bytes
- No header length field
- Uses payload length instead

Fixed header format

Offsets	Octet	0								1								2								3							
Octet	Bit	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
0	0	*Version*				*Traffic Class*								*Flow Label*																			
4	32	*Payload Length*																*Next Header*								*Hop Limit*							
8	64	*Source Address*																															
12	96																																
16	128																																
20	160																																
24	192	*Destination Address*																															
28	224																																
32	256																																
36	288																																

Routers process IPv6 faster due to simplified header.

IPv6 Header Fields Explanation

Version (4 bits)

Always set to 6 (binary 0110)

Traffic Class (8 bits)

Used for QoS to prioritize traffic
Example: VoIP, video calls

Flow Label (20 bits)

Identifies specific traffic flows between endpoints

Payload Length (16 bits)

Length of Layer 4 data in bytes
IPv6 header length is not included (fixed 40 bytes)

Next Header (8 bits)

Indicates encapsulated protocol (TCP/UDP)
Same purpose as IPv4 Protocol field

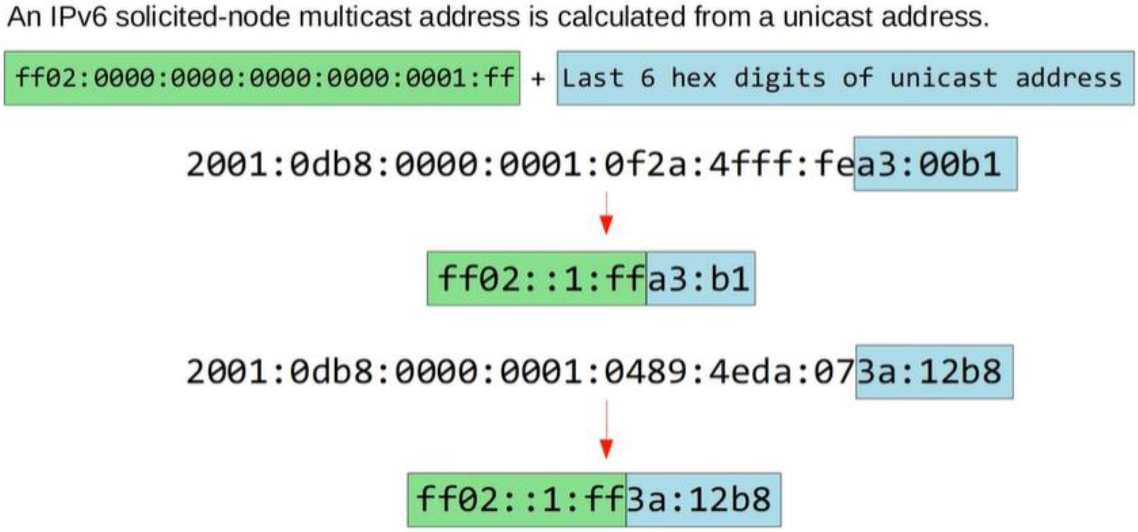
Hop Limit (8 bits)

Decreases by 1 per router
Packet is dropped when value reaches 0
Same role as IPv4 TTL

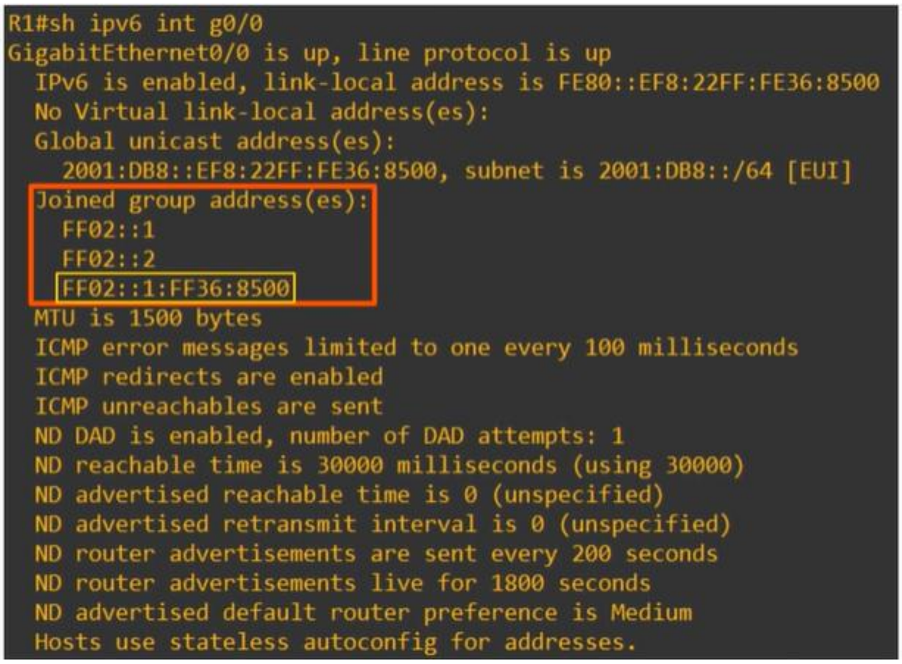
Source Address (128 bits)

Destination Address (128 bits)

IPv6 Solicited-Node Multicast Address



Router Multicast Group Membership:



```
R1#sh ipv6 int g0/0
GigabitEthernet0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::EF8:22FF:FE36:8500
  No Virtual link-local address(es):
  Global unicast address(es):
    2001:DB8::EF8:22FF:FE36:8500, subnet is 2001:DB8::/64 [EUI]
  Joined group address(es):
    FF02::1
    FF02::2
    FF02::1:FF36:8500
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND advertised reachable time is 0 (unspecified)
  ND advertised retransmit interval is 0 (unspecified)
  ND router advertisements are sent every 200 seconds
  ND router advertisements live for 1800 seconds
  ND advertised default router preference is Medium
  Hosts use stateless autoconfig for addresses.
```

Joined multicast groups:

FF02::1

FF02::2

FF02::1:FF36:8500 <-- Solicited-Node Address

Neighbour Discovery Protocol (NDP)

How NDP Replace ARP?

NDP replaces ARP in IPv6.

It uses:

- ICMPv6
- Solicited-node multicast

It is essential for IPv6 routing.

ARP vs NDP Difference

IPv4 ARP → Broadcast
IPv6 NDP → Multicast (more efficient)

NDP Message Types

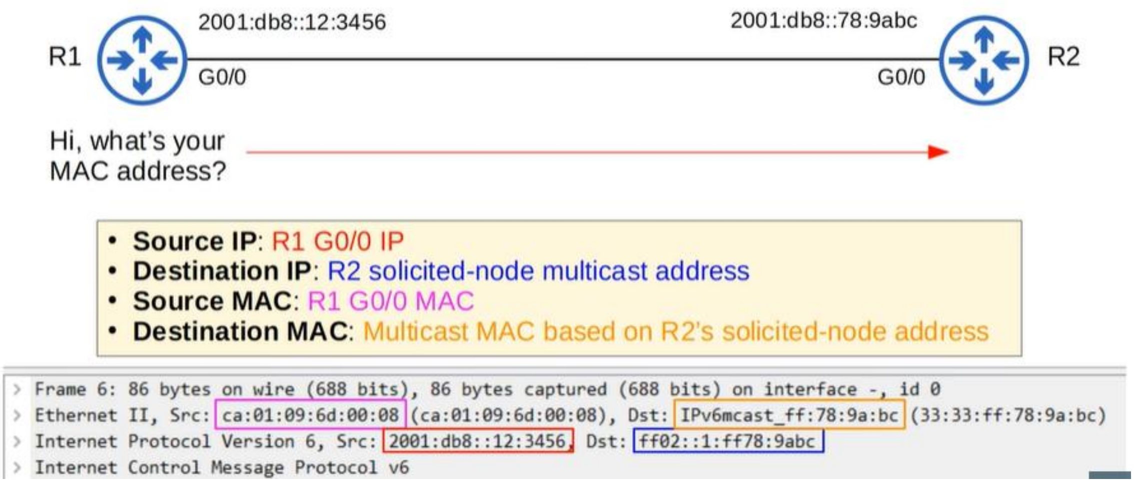
1] Neighbour Solicitation

- ICMPv6 Type 135
- Equivalent to ARP Request

2] Neighbour Advertisement

- ICMPv6 Type 136
- Equivalent to ARP Reply

Neighbour Solicitation Example



Neighbour Advertisement Example

- **Source IP**: R2 G0/0 IP
- **Destination IP**: R1 G0/0 IP
- **Source MAC**: R2 G0/0 MAC
- **Destination MAC**: R1 G0/0 MAC

```
> Frame 7: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface -, id 0
> Ethernet II, Src: ca:02:09:7c:00:08 (ca:02:09:7c:00:08), Dst: ca:01:09:6d:00:08 (ca:01:09:6d:00:08)
> Internet Protocol Version 6, Src: 2001:db8::78:9abc, Dst: 2001:db8::12:3456
> Internet Control Message Protocol v6
```

IPv6 Neighbour Table

Link layer -> Mac address


```
R1#show ipv6 neighbor
IPv6 Address                               Age Link-layer Addr State Interface
FE80::C802:9FF:FE7C:8                        0 ca02.097c.0008  REACH Gi0/0
2001:DB8::78:9ABC                            0 ca02.097c.0008  REACH Gi0/0
```

```
R2#show ipv6 neighbor
IPv6 Address                               Age Link-layer Addr State Interface
FE80::C801:9FF:FE6D:8                        0 ca01.096d.0008  REACH Gi0/0
2001:DB8::12:3456                            0 ca01.096d.0008  REACH Gi0/0
```

Router Discovery in NDP

- Another function of NDP allows hosts to automatically discover routers on the local network.
- Two messages are used for this process:

1) Router Solicitation (RS) = ICMPv6 Type 133
→ Sent to multicast address FF02::2 (all routers).
→ Asks all routers on the local link to identify themselves.
→ Sent when an interface is enabled/host is connected to the network.

2) Router Advertisement (RA) = ICMPv6 Type 134
→ Sent to multicast address FF02::1 (all nodes).
→ The router announces its presence, as well as other information about the link.
→ These messages are sent in response to RS messages.
→ They are also sent periodically, even if the router hasn't received an RS.

RS and RA Flow

SLAAC

Stateless Address Auto-Configuration

SLAAC stands for Stateless Address Auto-configuration.
It is another method to configure IPv6 addresses.

When using SLAAC:

- Hosts use RS and RA messages to learn the IPv6 prefix of the local link
  Example prefix: 2001:db8::/64
- Hosts then use that prefix to automatically generate an IPv6 address

When using the IPv6 address eui-64 command:

- The prefix must be manually entered

When using the IPV6 ADDRESS AUTOCONFIG command:

- The prefix does NOT need to be entered

This works because:

- NDP learns the prefix via RS and RA messages
- The device generates the Interface ID using EUI-64
- Or it may generate it randomly, depending on the vendor

SLAAC on Cisco Router

```
R2(config)#int g0/0
R2(config-if)#ipv6 address autoconfig
R2(config-if)#do show ipv6 interface brief
GigabitEthernet0/0     [up/up]
    FE80::EF8:22FF:FE56:A600
    2001:DB8::EF8:22FF:FE56:A600
GigabitEthernet0/1     [administratively down/down]
    unassigned
GigabitEthernet0/2     [administratively down/down]
    unassigned
GigabitEthernet0/3     [administratively down/down]
    unassigned
```

GigabitEthernet0/0   [up/up]

FE80::EF8:22FF:FE56:A600

2001:DB8::EF8:22FF:FE56:A600

Other interfaces:

Administratively down

R2 automatically generated:

- A global unicast address
- A link-local address

This behaviour is not Cisco-specific.
SLAAC is a standard IPv6 feature, and end hosts such as PCs can also use it.

Duplicate Address Detection (DAD)

Duplicate Address Detection (DAD) is another function of NDP.

DAD allows hosts to check if another device is using the same IPv6 address.

DAD runs when:

- An IPv6-enabled interface initializes (e.g., NO SHUTDOWN)
- An IPv6 address is configured manually
- An IPv6 address is configured via SLAAC

DAD uses:

- Neighbour Solicitation (NS)
- Neighbour Advertisement (NA)

No new message types are required.

DAD Process

To perform DAD:

- The host sends an NS to its own IPv6 address
- The destination is its own solicited-node multicast address

If no reply is received:

- The address is unique

If a Neighbour Advertisement reply is received:

- Another device is already using the address

On Cisco routers:

- A system message appears if a duplicate address is detected

```
*Oct 31 11:28:48.318: %IPV6_ND-4-DUPLICATE: Duplicate address 2001:DB8::1 on GigabitEthernet0/0
```

Further behaviour depends on:

- How the address was configured
- Device vendor
  (No deeper detail required for CCNA.)

IPv6 Static Routing

How is Different or Same as IPv6?

Now the topic moves to IPv6 static routing.

For CCNA:

- You must configure and verify the same types of static routes as IPv4

IPv6 routing works the same way as IPv4 routing:

- Router receives a packet
- Looks up destination in the routing table
- Forwards based on the most specific match

IPv4 routing and IPv6 routing:

- Operate separately
- Use separate routing tables

You can view IPv6 routing table with:

"SHOW IPV6 ROUTE"

IPv6 Routing Enablement

IPv4 routing is enabled by default.

IPv6 routing is disabled by default and must be enabled with:

"IPV6 UNICAST-ROUTING"

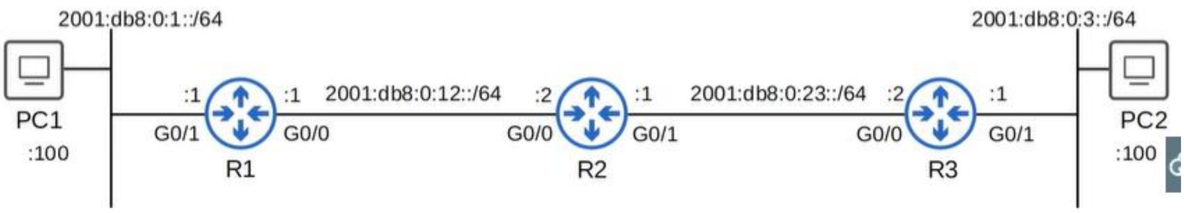
If IPv6 routing is disabled:

- The router can send/receive IPv6
- But will not forward traffic between networks

If IPv6 seems correct but networking fails:

- You likely forgot IPV6 UNICAST-ROUTING

Network Topology Used



IPv6 Routing Table — Connected & Local Routes

```
R1#show ipv6 route
IPv6 Routing Table - default - 5 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP
       H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
       IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       RL - RPL, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       la - LISP alt, lr - LISP site-registrations, ld - LISP dyn-eid
       lA - LISP away, a - Application
C   2001:DB8:0:1::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2001:DB8:0:1::1/128 [0/0]
     via GigabitEthernet0/1, receive
C   2001:DB8:0:12::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:0:12::1/128 [0/0]
     via GigabitEthernet0/0, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

Observations:

- Connected network routes are added automatically
- Local host routes (/128) are added automatically
- FF00::/8 (multicast range) is discarded via Null0
- Link-local routes are NOT added to routing table

IPv6 Static Route Command Format

```
ipv6 route destination/prefix-length {next-hop | exit-interface [next-hop]} [ad]
```

Meaning of Symbols

- Curly braces {} → Required choice
- Square brackets [] → Optional
- AD → Administrative Distance
- AD is used for floating static routes.

Types of IPv6 Static Routes

1. Directly Attached Static Route

Only the exit interface is specified.

```
ipv6 route destination/prefix-length exit-interface
  R1(config)# ipv6 route 2001:db8:0:3::/64 g0/0
```

2. Recursive Static Route

Only the next-hop address is specified.

```
ipv6 route destination/prefix-length next-hop
  R1(config)# ipv6 route 2001:db8:0:3::/64 2001:db8:0:12::2
```

It is called recursive because:

- Router looks up destination
- Then looks up next-hop
- Then determines exit interface

Recursive Lookup Flow:

Packet -> Match Route -> Next Hop -> Lookup Next Hop -> Select Interface

3. Fully Specified Static Route

Both exit interface and next hop are specified.

```
ipv6 route destination/prefix-length exit-interface next-hop
  R1(config)# ipv6 route 2001:db8:0:3::/64 g0/0 2001:db8:0:12::2
```

IPv6 Directly Attached Route Limitation

In IPv6: Directly attached static routes DO NOT work on Ethernet interfaces

Even if the command is accepted: The router will not forward traffic

They only work on non-Ethernet interfaces (e.g., serial)

Therefore: Use recursive or fully specified routes instead

Directly attached static route: Only the exit interface is specified.
`ipv6 route destination/prefix-length exit-interface`
~~R1(config)# ipv6 route 2001:db8:0:3::/64 g0/0~~

In IPv6, you CAN'T use directly attached static routes if the interface is an Ethernet interface.

IPv6 Static Route Examples

Network Route (on R1)

```
R1(config)# ipv6 route 2001:db8:0:3::/64 2001:db8:0:12::2
```

Tells R1 to send traffic to R2

Host Routes (on R2)

```
R2(config)# ipv6 route 2001:db8:0:1::100/128 2001:db8:0:12::1
R2(config)# ipv6 route 2001:db8:0:3::100/128 2001:db8:0:23::2
```

Uses /128 prefix for single-host routing

Default Route (on R3)

```
R3(config)# ipv6 route ::/0 2001:db8:0:23::1
```

Equivalent to IPv4 0.0.0.0/0

"All Routes Shown Are Recursive They specify only next hop"

Floating Static Routes

Floating static routes are created by raising Administrative Distance (AD).

Static route AD must be:

- Higher than OSPF (110)
- Higher than EIGRP (90)

This ensures it is used only as backup

Link-Local Address as Next-Hop

Attempting to use a link-local next-hop alone causes error:

Interface has to be specified for a link-local nexthop

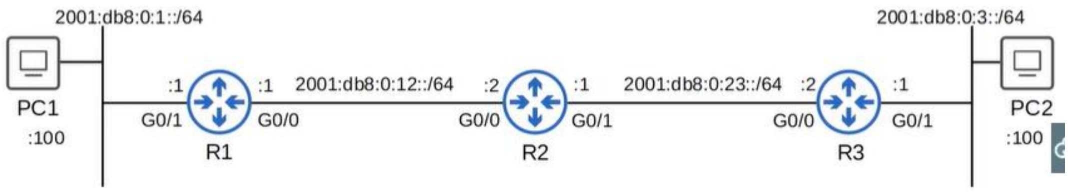
To use link-local next-hop:

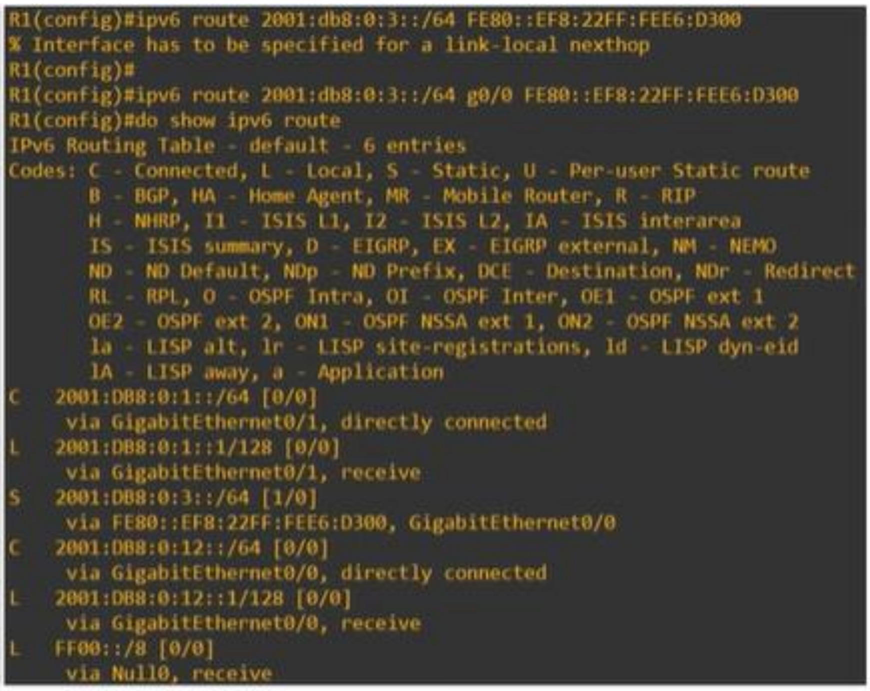
- Must specify both next hop and exit interface

This becomes a fully specified static route

Reason:

- Router cannot determine the correct interface automatically





```
R1(config)#ipv6 route 2001:db8:0:3::/64 FE80::EF8:22FF:FEE6:D300
% Interface has to be specified for a link-local nexthop
R1(config)#
R1(config)#ipv6 route 2001:db8:0:3::/64 g0/0 FE80::EF8:22FF:FEE6:D300
R1(config)#do show ipv6 route
IPv6 Routing Table - default - 6 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
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       H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
       IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
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       la - LISP alt, lr - LISP site-registrations, ld - LISP dyn-eid
       lA - LISP away, a - Application
C   2001:DB8:0:1::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2001:DB8:0:1::1/128 [0/0]
     via GigabitEthernet0/1, receive
S   2001:DB8:0:3::/64 [1/0]
     via FE80::EF8:22FF:FEE6:D300, GigabitEthernet0/0
C   2001:DB8:0:12::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:0:12::1/128 [0/0]
     via GigabitEthernet0/0, receive
L   FF00::/8 [0/0]
     via Null0, receive
```